VF Asia Footwear Laboratory Testing Manual

Test Method for Thickness / Gauge	**Test method # :VF-A03** **REF #: TM-R- 015** **Page 1 of 3**
Date Issued: April , 2014 **Revision #: 2** **Reason for revision: TBL and VF aligned the test method. For laboratory test – should follow the test specified per material. In the production line may use single method with the use of calipers or thickness gauge.**	**Created by: Nila Babia Erazo**

PURPOSE AND SCOPE: To measure the thickness or gauge of a test material. This method is applicable with any type of flat material.

REFERENCES: ASTM D1813, ASTM D1777

EQUIPMENT AND MATERIAL:

1. Thickness gauge as described below in sections A, and B.
2. Calipers capable of measuring up to 0.01mm. Digital type is preferred.
3. Leather, synthetic leather, plastic, fabric, rubber, foam, thread, shoe lace thread, shoe lace, paper/fiber board, etc.

A) Leather, synthetic leather and plastics. (ASTM D1813)

To determine the overall thickness (range) of a test material the thickness gauge to be used must conform to the following:

1. The dead mass loading is equivalent to 393 ± 10g (13.86± 0.35ounces).
2. The presser foot is 10.0 ± **0.5** mm in diameter.
3. The surface of the base plate is parallel to the surface of the presser foot.

Note: The overall loading is equivalent to 500 g/cm^2.

B) Fabric, rubber, foam, thread, reinforcements and paper/fiber board. (ASTM D1777)

To determine the overall thickness (range) of a test material the thickness gauge must conform to the following:

1. The dead mass loading is equivalent to 60 ± 5g (2.0 ± 0.3 ounces).
2. The presser foot is 10.0 ± **0.5** mm in diameter
3. The surface of the base plate is parallel to the surface of the presser foot

Note: The overall loading is equivalent to 70g/cm^2.

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C) Multi-layer fabric and shoe lace.

Laces and multi-layer fabrics should be measured using Vernier calipers.

SAMPLE PREPARATION:

Condition the sample at standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity prior to test.

REPETITIONS: Take at least six measurements over the entire surface of the test material, or measure as specified in the **VF Asia** test method requiring the thickness measurement(s). For threads there should be 10 meters between measurements

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A）GENERAL TEST PROCEDURE:

1. To determine the overall thickness (average/range) of a test material, use the entire test material. To determine the thickness of a test sample, prepare the test sample as described in the **VF Asia** test method being performed.
2. Raise the presser foot high or open the calipers enough so that the test sample can be inserted.
3. Place the test sample on the base plate such that it rests flush on the surface without wrinkles.
4. Gently lower (do not drop) the presser foot or close the calipers onto the test sample.
5. Wait five seconds and read the thickness to the nearest 0.01mm.

REPORT: The range and the average of the thickness measurements to the nearest 0.01mm

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B) PU COATING THICKNESS TEST PROCEDURE:

1. Cut a thin cross-section of the leather (approximately 2-3mm wide by 3cm long).
   NOTE: *Test samples must have sides that are perpendicular to the top and bottom of the leather.*
2. Place the test sample on the base of a microscope, such that the cross-section of the leather can be viewed.
3. Using the appropriate measuring attachment for the microscope, measure the thickness of the PU coating on the leather to the nearest 0.01mm.
4. Take 3 measurements from each of the two test samples.

REPORT: The range and the average of the thickness measurements to the nearest 0.01mm.

C) MULTI-LAYER FABRIC AND SHOE LACE TEST PROCEDURE

1. To determine the thickness of a test sample of a multi-layer fabric, prepare the test sample as described in the specific **VF Asia** test method being performed .For a shoe lace use the whole lace.
2. Place the test sample on the between the jaws of the Vernier calipers.
3. Gently lower close the jaws of the Vernier calipers until they just make contact with the fabric or shoe lace.
4. Take 3 measurements from each of the two test samples.
5. Read the thickness to the nearest 0.01mm.

REPORT: The range and the average of the thickness measurements to the nearest 0.01mm.